

Set 11

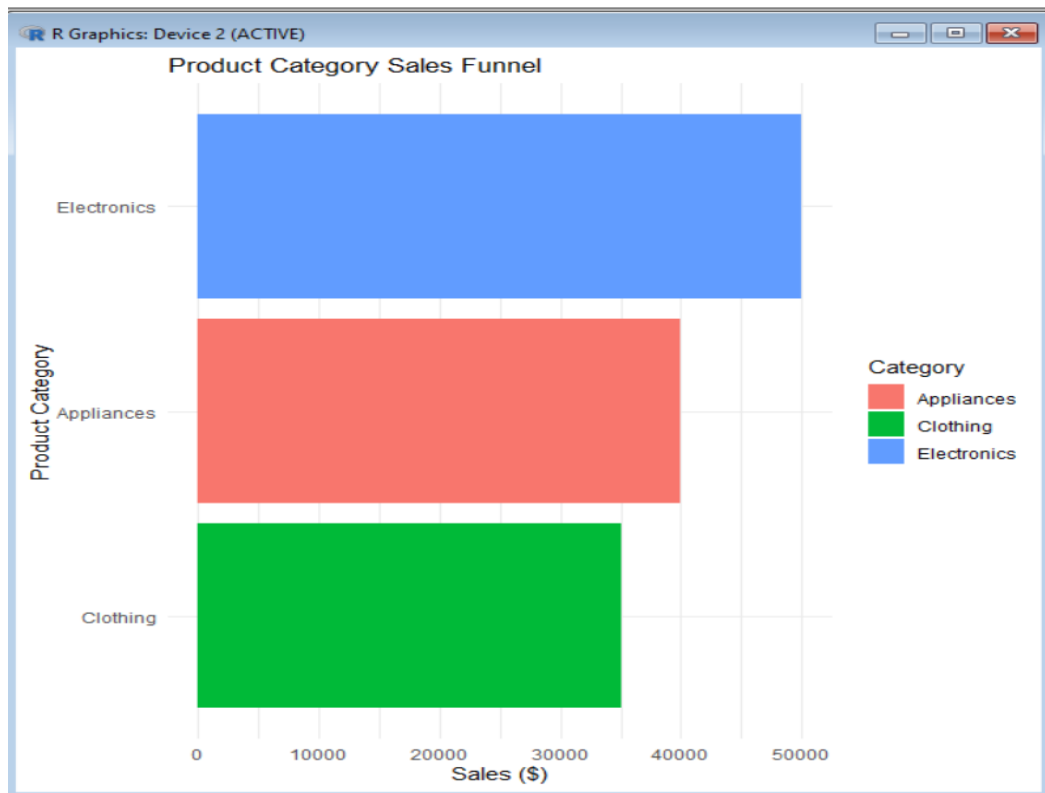
1. funnel chart to analyze the sales conversion process for each product category:

python code:

```
product <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```
Product  
ggplot(product,  
  aes(x = reorder(Category, Sales),  
    y = Sales,  
    fill = Category)) +  
geom_col() +  
coord_flip() +  
labs(  
  title = "Product Category Sales Funnel",  
  x = "Product Category",  
  y = "Sales ($)"  
) +  
theme_minimal()
```

Output:

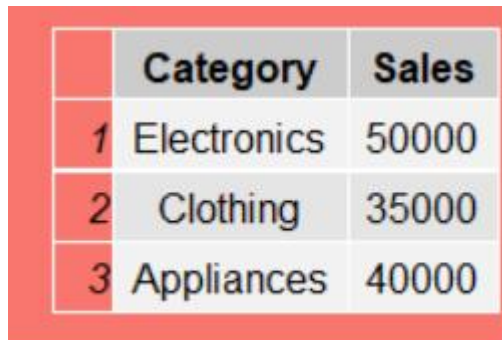


2. table to display the sales data for each product category:

python code:

```
grid.table(product)
```

output:



	Category	Sales
1	Electronics	50000
2	Clothing	35000
3	Appliances	40000

3. dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

Python code:

Pie Chart

```
p1 <- ggplot(product,
  aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  geom_text(aes(label = paste0(round(Sales/sum(Sales)*100), "%")),
    position = position_stack(vjust = 0.5),
    color = "white") +
  labs(title = "Sales Distribution") +
  theme_void()
```

Funnel Chart

```
p2 <- ggplot(product,
  aes(x = reorder(Category, Sales),
    y = Sales,
    fill = Category)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel",
    x = "Category",
    y = "Sales"
  )
```

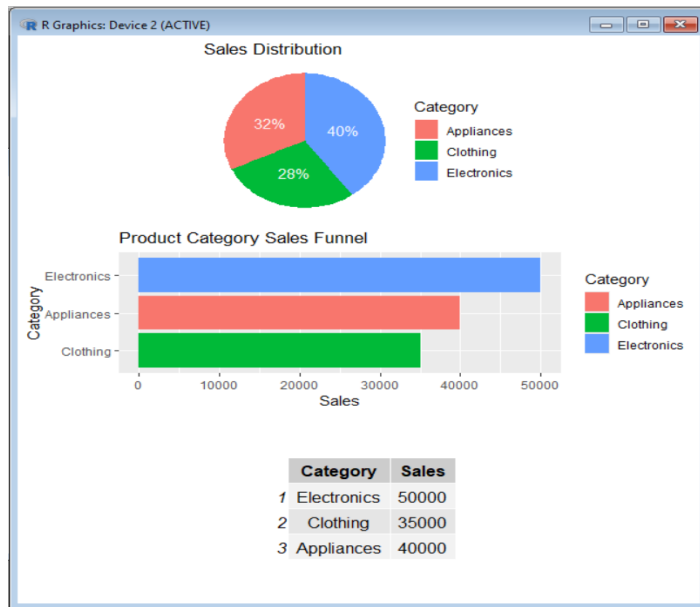
Dashboard

```

grid.arrange(
  p1,
  p2,
  tableGrob(product),
  ncol = 1
)

```

Output:



4. pie chart to represent the distribution of sales across product categories.

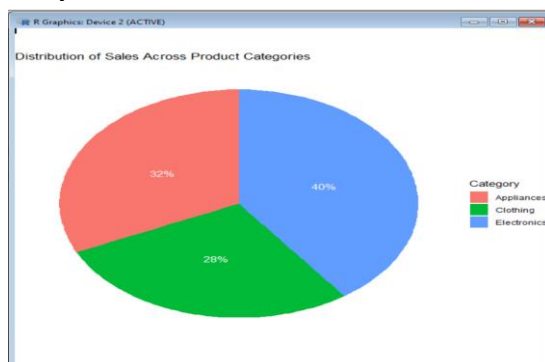
Python code:

```

ggplot(product,
  aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  geom_text(aes(label = paste0(round(Sales/sum(Sales)*100), "%")),
    position = position_stack(vjust = 0.5),
    color = "white") +
  labs(title = "Distribution of Sales Across Product Categories") +
  theme_void()

```

Output:



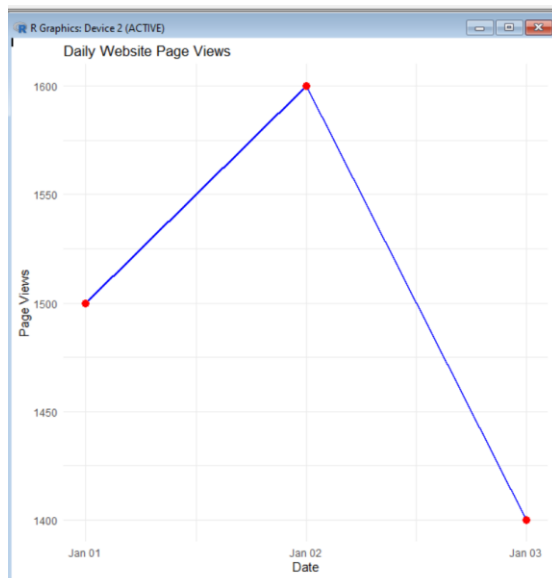
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1. line chart to visualize the trend in daily page views over time.

Python code:

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  PageViews = c(1500, 1600, 1400),  
  CTR = c(2.3, 2.7, 2.0)  
)  
  
traffic  
ggplot(traffic,  
  aes(x = Date,  
    y = PageViews,  
    group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Website Page Views",  
    x = "Date",  
    y = "Page Views"  
  ) +  
  theme_minimal()
```

Output:



2. bar chart showing the top N days with the highest click-through rates.

Python code:

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
```

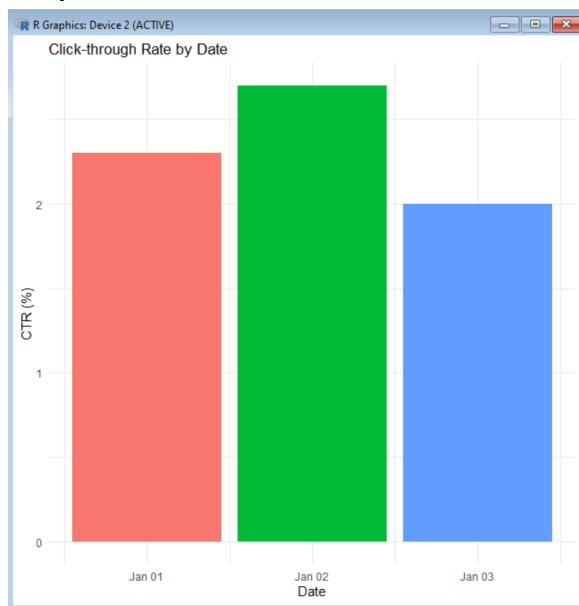
```

PageViews = c(1500,1600,1400),
CTR = c(2.3,2.7,2.0)
)

traffic
ggplot(traffic,
  aes(x = Date,
      y = CTR,
      fill = factor(Date))) +
geom_col() +
labs(
  title = "Click-through Rate by Date",
  x = "Date",
  y = "CTR (%)"
) +
theme_minimal() +
theme(legend.position = "none")

```

Output:



3. table to show the website traffic data for each day.

Python code:

```
grid.table(traffic)
```

Output:

	Date	PageViews	CTR
1	2023-01-01	1500	2.3
2	2023-01-02	1600	2.7
3	2023-01-03	1400	2.0

4. dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Python code:

```
p1 <- ggplot(traffic,
             aes(x = Date,
                 y = PageViews,
                 group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Website Page Views",
    x = "Date",
    y = "Page Views"
  )
```

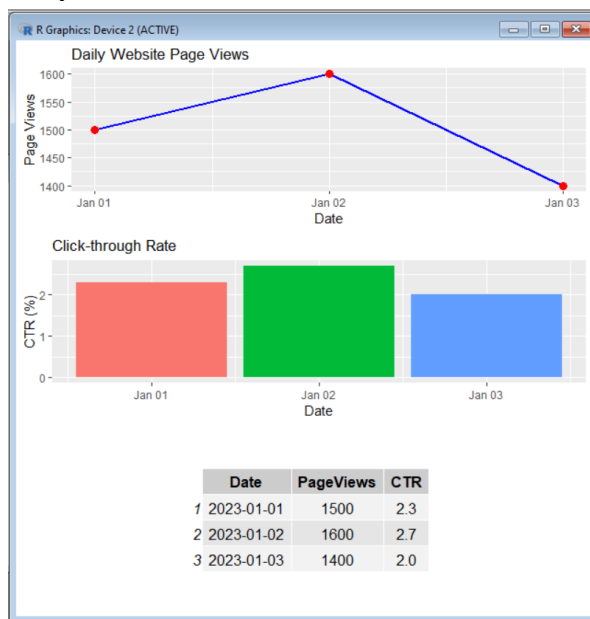
Bar Chart

```
p2 <- ggplot(traffic,
             aes(x = Date,
                 y = CTR,
                 fill = factor(Date))) +
  geom_col() +
  labs(
    title = "Click-through Rate",
    x = "Date",
    y = "CTR (%)"
  ) +
  theme(legend.position = "none")
```

Dashboard

```
grid.arrange(
  p1,
  p2,
  tableGrob(traffic),
  ncol = 1
)
```

Output:



Set 13

1. map chart to visualize the distribution of cities on a geographic map.

python code:

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  Temperature = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200),  
  Longitude = c(78.48, 77.59, 80.27),  
  Latitude = c(17.38, 12.97, 13.08)  
)
```

```
geo  
library(ggplot2)  
library(maps)
```

```
world <- map_data("world")
```

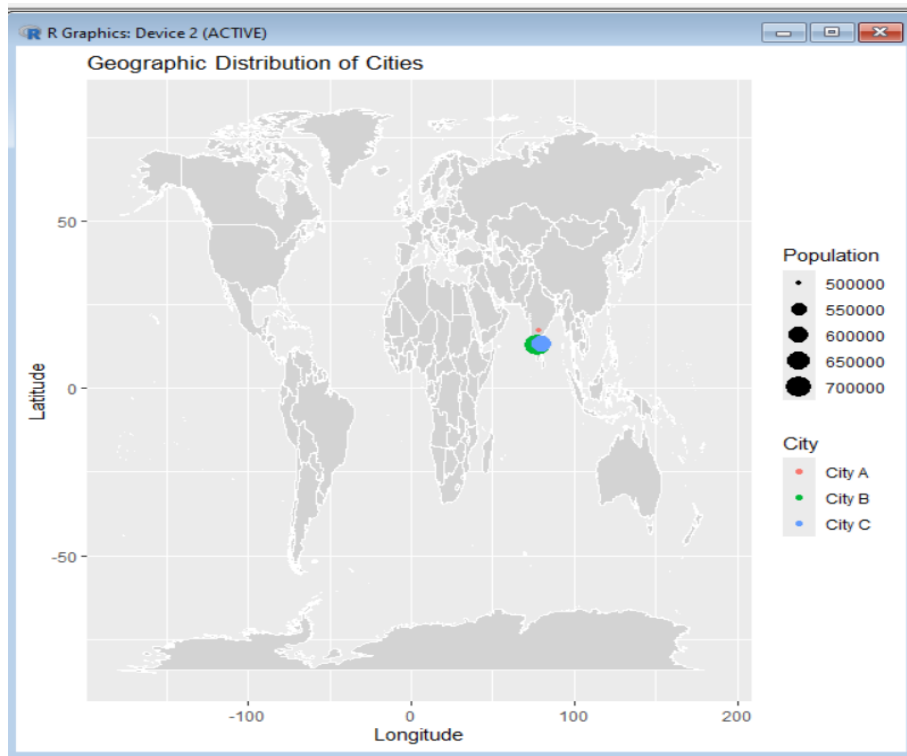
```
ggplot() +  
  geom_map(data = world,  
    map = world,  
    aes(long, lat, map_id = region),  
    fill = "lightgray",  
    color = "white") +  
  geom_point(data = geo,  
    aes(x = Longitude,
```

```

    y = Latitude,
    size = Population,
    color = City)) +
labs(
  title = "Geographic Distribution of Cities",
  x = "Longitude",
  y = "Latitude"
)

```

output:



2. scatter plot to explore the relationship between average temperature and population size.

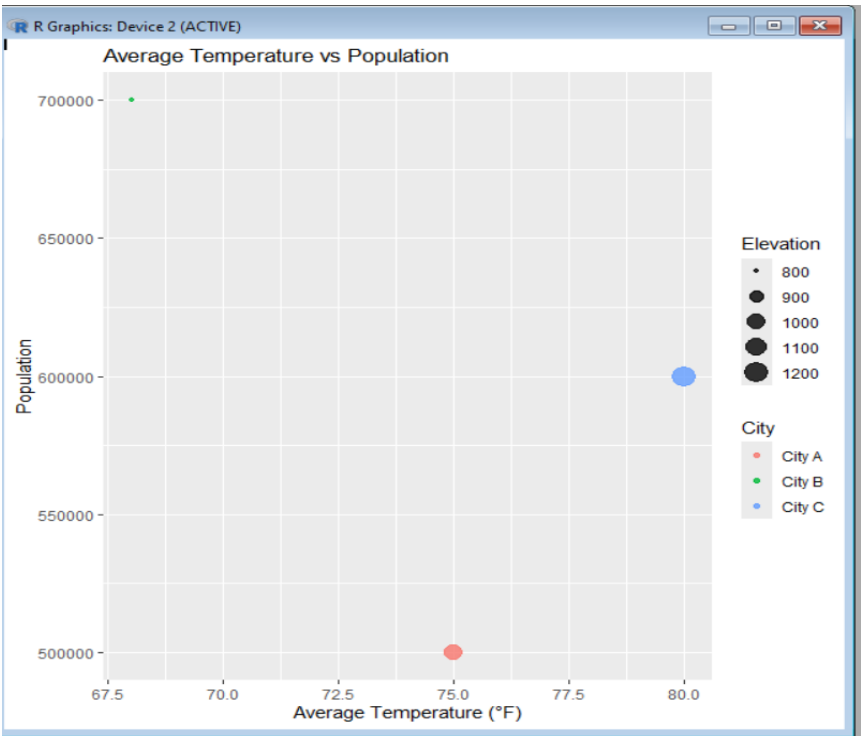
Python code:

```

ggplot(geo,
  aes(x = Temperature,
    y = Population,
    color = City,
    size = Elevation)) +
geom_point(alpha = 0.8) +
labs(
  title = "Average Temperature vs Population",
  x = "Average Temperature (°F)",
  y = "Population"
)

```


Output:



3. table to display the geographic data for each city.

Python code:

```
grid.table(gео)
```

Output:

	City	Population	Temperature	Elevation	Longitude	Latitude
1	City A	5e+05	75	1000	78.48	17.38
2	City B	7e+05	68	800	77.59	12.97
3	City C	6e+05	80	1200	80.27	13.08

4. dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Python code:

```
# Map Chart
world <- map_data("world")

p1 <- ggplot() +
  geom_map(data = world,
    map = world,
    aes(long, lat, map_id = region),
    fill = "lightgray",
    color = "white") +
  geom_point(data = geo,
```

```

aes(Longitude,
     Latitude,
     size = Population,
     color = City)) +
labs(title = "City Map")

```

Scatter Plot

```

p2 <- ggplot(geo,
             aes(Temperature,
                 Population,
                 color = City,
                 size = Elevation)) +
geom_point() +
labs(title = "Temperature vs Population")

```

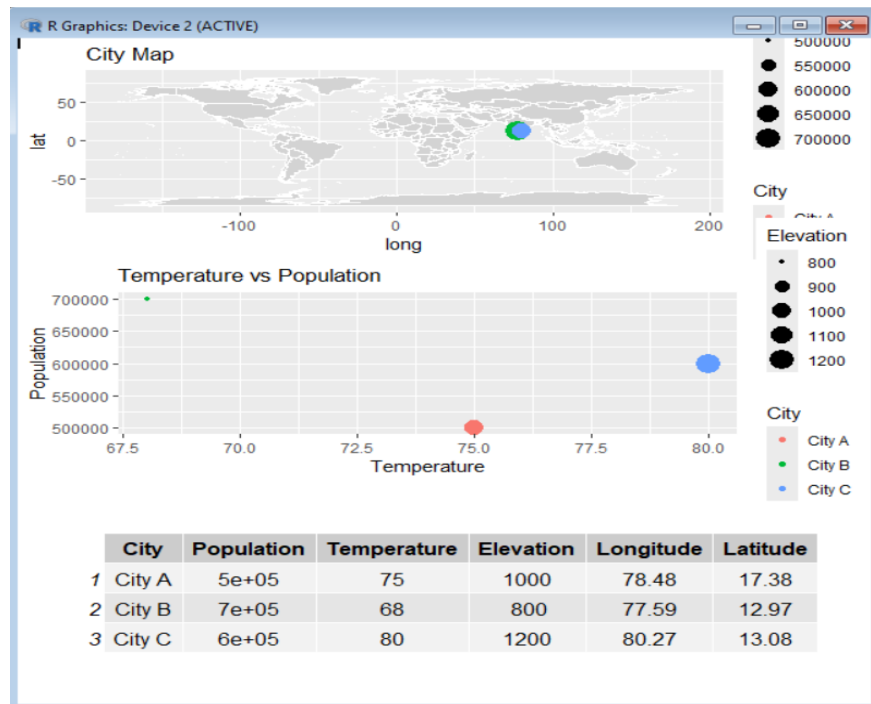
Dashboard

```

grid.arrange(
  p1,
  p2,
  tableGrob(geo),
  ncol = 1
)

```

Output:



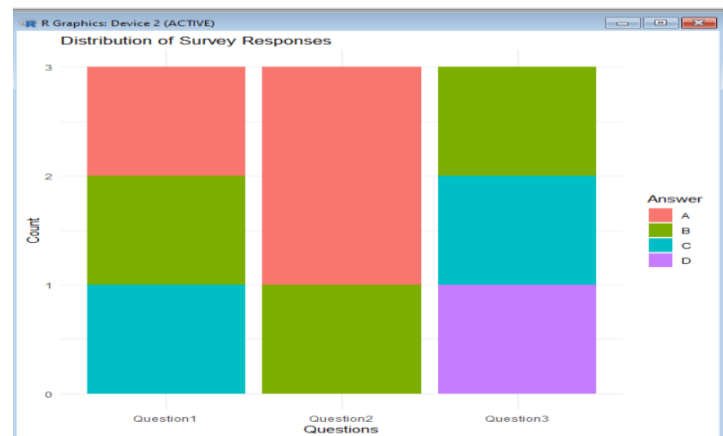
Set 14

1. stacked bar chart to display the distribution:

python code:

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)  
  
survey  
library(tidyr)  
  
survey_long <- pivot_longer(  
  survey,  
  cols = c(Question1, Question2, Question3),  
  names_to = "Question",  
  values_to = "Answer"  
)  
  
ggplot(survey_long,  
  aes(x = Question,  
    fill = Answer)) +  
  geom_bar() +  
  labs(  
    title = "Distribution of Survey Responses",  
    x = "Questions",  
    y = "Count"  
  ) +  
  theme_minimal()
```

Output:



2. radar chart to visualize the overall pattern of responses

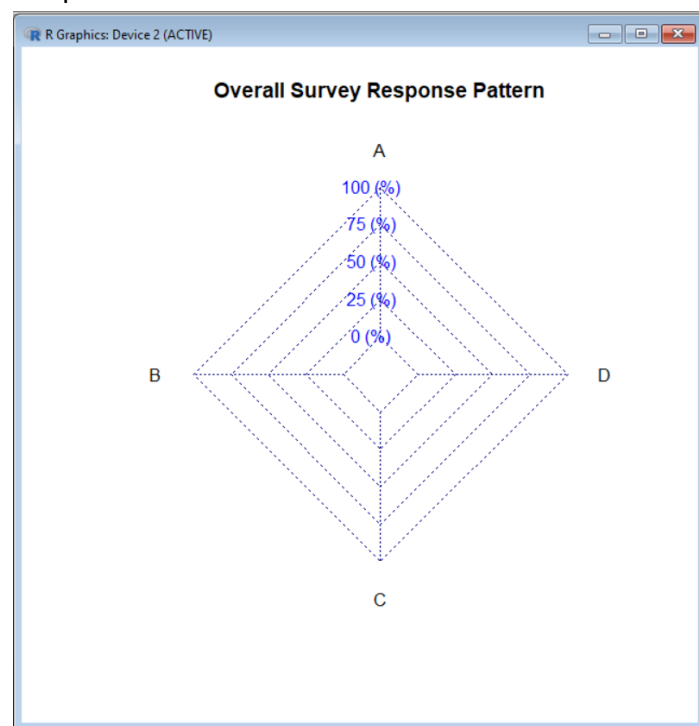
python code:

```
radar <- data.frame(  
  A = c(3,1),  
  B = c(3,1),  
  C = c(3,1),  
  D = c(3,0)  
)
```

```
rownames(radar) <- c("Max", "Responses")
```

```
radarchart(  
  radar,  
  axistype = 1,  
  pcol = "blue",  
  pfcpl = rgb(0,0,1,0.3),  
  plwd = 2,  
  title = "Overall Survey Response Pattern"  
)
```

Output:



3. table to show the survey response data for each respondent.

Python code:

```
grid.table(survey)
```

Output:

	Respondent	Question1	Question2	Question3	
1	1	A	B	C	D
2	2	B	A	D	
3	3	C	A	B	

4. dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Python code:

```
p1 <- ggplot(survey_long,
             aes(x = Question,
                 fill = Answer)) +
  geom_bar() +
  labs(title = "Stacked Bar Chart")
```

Table

```
tbl <- tableGrob(survey)
```

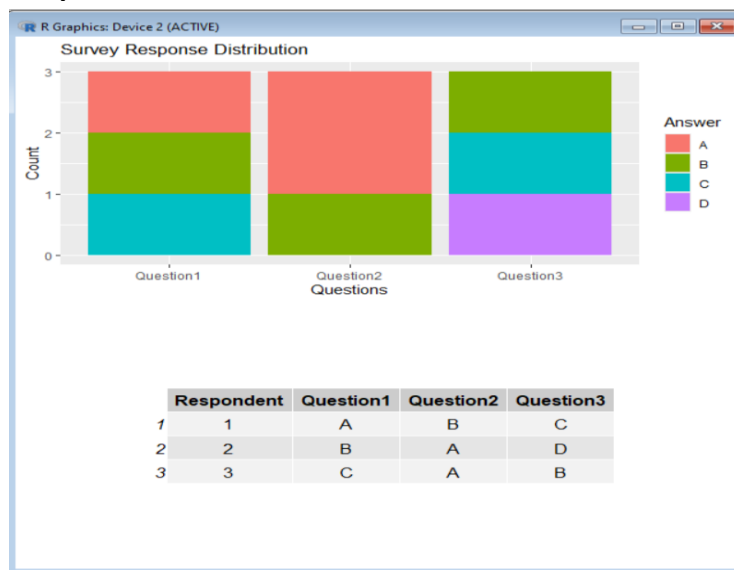
Display Dashboard

```
grid.arrange(
  p1,
  tbl,
  ncol = 1
)
```

Run the radar chart separately

```
radarchart(
  radar,
  axistype = 1,
  pcol = "blue",
  pfc = rgb(0,0,1,0.3),
  plwd = 2,
  title = "Overall Survey Response Pattern"
)
```

Output:



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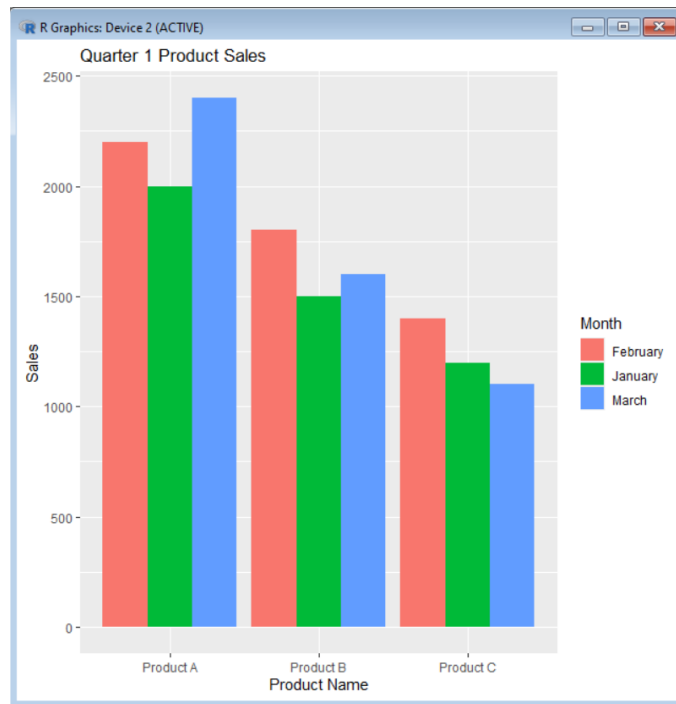
1. grouped bar chart to visualize the sales of each product for the first quarter.

Python code:

```
sales <- data.frame(  
  ProductID = c(1,2,3),  
  ProductName = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
sales_long <- pivot_longer(  
  sales,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)  
  
ggplot(sales_long,  
  aes(x = ProductName,  
    y = Sales,  
    fill = Month)) +  
  geom_bar(stat = "identity",  
    position = "dodge") +  
  labs(  
    title = "Quarter 1 Product Sales",  
    x = "Product Name",
```

```
y = "Sales"
)
```

Output:



2. **stacked area chart to represent the overall sales trend for all products over the three months.**

Python code:

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = ProductName,
    group = ProductName)) +
  geom_area(alpha = 0.8) +
```

```

labs(
  title = "Overall Sales Trend",
  x = "Month",
  y = "Sales"
)

```

Output:



3. table to show the monthly sales data for each product.

Python code:

```
grid.table(sales)
```

Output:

						ProductNa
	ProductID	ProductName	January	February	March	
1	1	Product A	2000	2200	2400	Product
2	2	Product B	1500	1800	1600	Product
3	3	Product C	1200	1400	1100	Product

4. dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Python code:

```
library(gridExtra)
```

```

p1 <- ggplot(sales_long,
  aes(x = ProductName,

```



```

      y = Sales,
      fill = Month)) +
geom_bar(stat = "identity",
         position = "dodge") +
labs(
  title = "Grouped Bar Chart",
  x = "Product",
  y = "Sales"
)

p2 <- ggplot(sales_long,
             aes(x = Month,
                 y = Sales,
                 fill = ProductName,
                 group = ProductName)) +
geom_area(alpha = 0.8) +
labs(
  title = "Stacked Area Chart",
  x = "Month",
  y = "Sales"
)

grid.arrange(
  p1,
  p2,
  tableGrob(sales),
  ncol = 1
)

```

Output:

